

SentinelNet: Professional Documentation for AI-Powered Network Intrusion Detection System

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Abstract

SentinelNet is a comprehensive AI-based Network Intrusion Detection System (NIDS) developed for accurate, real-time detection and classification of malicious network traffic. Leveraging the CICIDS 2017 and NSL-KDD datasets, the system employs a hybrid ensemble model combining Random Forest and Gradient Boosting algorithms. This design ensures exceptional accuracy, low false-positive rates, and robust performance across a wide range of attack types, making it a scalable solution for modern network security needs.

1. Introduction

The surge in sophisticated cyber threats necessitates advanced network security solutions. Traditional intrusion detection methods often fall short due to limited adaptability and lower detection rates. SentinelNet addresses these challenges by applying machine learning techniques to detect anomalies and malicious activities efficiently, thereby strengthening network security infrastructure.

2. Literature Review

Extensive research demonstrates that machine learning approaches such as Random Forest, Support Vector Machines (SVM), and Gradient Boosting enhance NIDS performance. Ensemble methods, which combine multiple classifiers, improve accuracy, reduce false positives, and provide better generalization to unseen attacks. SentinelNet integrates these approaches to establish a highly effective detection framework.

3. Methodology

3a. Datasets

The system employs the following benchmark datasets:

Dataset	Samples	Features	Classes
CICIDS 2017	2,830,000+	78	Normal / Attack
NSL-KDD	125,973	41	Normal / Attack

3b. Data Preprocessing

- Cleaning and handling of inconsistent or missing data.
- Normalization and scaling of numerical features.
- Encoding categorical variables for model compatibility.

3c. Novel Contributions

- Hybrid ensemble model combining Random Forest and Gradient Boosting.
- Automated feature importance evaluation to highlight critical network traffic attributes.

3d. Proposed Workflow

1. Preprocess raw network traffic data.
2. Extract relevant features for classification.
3. Train ensemble classifiers on processed datasets.
4. Evaluate performance using accuracy, precision, recall, F1-score, and ROC-AUC.

4. Results

4a. Exploratory Data Analysis (EDA)

- Detailed analysis of network traffic distributions.
- Identification of key attack types in both datasets.

4b. Visualizations

- Confusion matrices, ROC-AUC curves, and feature importance plots generated for performance assessment.

4c. Performance Metrics

Dataset	Accuracy	Precision	Recall	F1-score
CICIDS 2017	99%	98.5%	98%	98.2%
NSL-KDD	97.5%	96.8%	95.7%	96.2%

4d. ROC-AUC Analysis

- Both datasets achieve ROC-AUC scores above 0.97, confirming strong discriminatory capability and reliable classification performance.

5. Discussion

SentinelNet demonstrates effective detection of multiple attack types with minimal false positives. The hybrid ensemble approach enhances model generalization and adaptability to new network threats. Feature importance evaluation provides actionable insights for network administrators to understand attack patterns and prioritize defenses.

6. Conclusion

SentinelNet proves the effectiveness of AI-based NIDS for proactive network security. The system achieves high detection accuracy, robust alerting, and provides actionable insights, validating its applicability for real-world network security scenarios using both CICIDS 2017 and NSL-KDD datasets.

7. Future Work

- Integrate deep learning models for advanced anomaly detection.
 - Develop a real-time monitoring and automated alert system.
 - Implement continuous retraining using updated network traffic data.
 - Expand capabilities to handle encrypted traffic and novel attack vectors.
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